

Quiz 2 — Cleaning and Counting

M 375T Mathematical Foundations of Machine Learning — Fall 2026

15 minutes · closed book · no devices · same skills as Practice 2 & 3

Name: _____ UT EID: _____

1. A feature has sample mean $\hat{\mu} = 6$ and sample standard deviation $\hat{\sigma} = 2$. We standardize it:
 $z = (x - \hat{\mu})/\hat{\sigma}$.

(i) What value does the data point $x = 8$ map to?

(ii) Which original value x maps to $z = -1.5$? [2 pts]

Answer:

2. Read the code, then answer in words and numbers:

```
U = rng.uniform(size=(50_000, 100))
k = (U < 0.5).sum(axis=1)
print(k.shape)
print(np.mean(k == 50))
```

(a) What does the first print statement output, and — in one sentence — what does each entry of k count?

(b) The second print estimates a probability. Which one? Give the exact expression for it (no need to evaluate). [4 pts]

Answer:

3. Quarter the interval into four bins. The model is uniform, $u = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, and $n = 100$ samples produce the observed histogram $q = (\frac{1}{2}, \frac{1}{2}, 0, 0)$. On paper:
- (a) Compute $D(q||u) = \sum_{\omega} q(\omega) \log \frac{q(\omega)}{u(\omega)}$ exactly (recall $0 \log 0 = 0$; a clean closed form exists).
 - (b) Using Sanov's estimate $\mathbb{P}(\text{histogram} \approx q) \approx e^{-nD(q||u)}$, express this probability as a power of 2. Should we keep the uniform model? [4 pts]

Answer:

Total: 10 points. Lowest two quiz scores are dropped at the end of the semester.